

Ji-Yeon Kim

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Research Interests

3D/4D Reconstruction and Generation of Humans, Objects, and Scenes

Monocular Reconstruction of Human and Object Motion; Collaborative Human–Object Interactions;
Text-Conditioned 3D Avatar Generation; Depth-Aware Video Segmentation

Education

Pohang University of Science and Technology, POSTECH

Integrated M.S. & Ph.D., Convergence of IT Engineering (Advisor: Prof. Tae-Hyun Oh)

Pohang, Korea

Mar. 2020 – Present

University of Toronto

Exchange Student (one semester), Faculty of Arts & Science

Toronto, Canada

Jul. 2018 – Jan. 2019

Sookmyung Women's University

Bachelor of Engineering, Major: IT Engineering

Summa Cum Laude; Ranked 1st in the College of Engineering

Seoul, Korea

Mar. 2015 – Aug. 2019

Experience

International Computer Vision Summer School (ICVSS)

Computer Vision for Spatial Intelligence

Sicily, Italy

Jul. 2025

Awards & Honors

AFCV Best Robot Vision Paper Award, KRoC

“Depth-aware Video Panoptic Segmentation Model for Holistic Scene Understanding”

Feb. 2024

Minister's Award (1st Place, \$12,000), Electronics Times ICT Paper Awards

“Text-driven 4D Human Avatar Generation via Cross-modal Synthesis-through-Optimization”

Dec. 2023

Qualcomm Innovation Fellowship Korea, Winner (\$4,000)

“CLIP-Actor: Text-Driven Recommendation and Stylization for Animating Human Meshes”

Nov. 2022

Outstanding Poster Presentation Award, IPIU

“Generating Action-Conditioned 3D Human in Scenes”

Feb. 2022

National Scholarship for Science and Engineering

Korea Student Aid Foundation

2017 – 2019

Academic Excellence Scholarship

Sookmyung Women's University

2015 – 2016

Publications

(Equal contribution are denoted by “*”).

Abbreviations

TPAMI	IEEE Transactions on Pattern Analysis and Machine Intelligence
RA-L	IEEE Robotics and Automation Letters
CVPR	IEEE/CVF Conference on Computer Vision and Pattern Recognition
ICCV	IEEE/CVF International Conference on Computer Vision
ECCV	European Conference on Computer Vision
ACCV	Asian Conference on Computer Vision
BMVC	British Machine Vision Conference
IROS	IEEE/RSJ International Conference on Intelligent Robots and Systems

Journal Articles

- [J03] K. Youwang*, T. Byun*, **Kim Ji-Yeon**, S. Choi, T.-H. Oh, “CLIP-Actor-X: Text-driven 4D Human Avatar Generation via Cross-modal Synthesis-through-Optimization.”
TPAMI 2026 (Minister’s Award, ICT Paper Awards)
- [J02] W. Jo, H. Ha, **Kim Ji-Yeon**, H. Jeong, T.-H. Oh, “LiDAR-Anchored Collaborative Distillation for Robust 2D Representations.”
RA-L 2026
- [J01] **Kim Ji-Yeon**, O. Hyun-Bin, K. Byung-Ki, D. Kim, Y. Kwon, T.-H. Oh, “Uni-DVPS: Unified Model for Depth-Aware Video Panoptic Segmentation.”
RA-L 2024 (Oral Presentation, IROS 2024)

Conference Papers

- [C09] W. Jo, H. Ha, **Kim Ji-Yeon**, H. Jeong, T.-H. Oh, “Self-Supervised Collaborative Distillation: Enhancing Lighting Robustness and 3D Awareness.”
ICCV 2025 Workshop on Wild3D
- [C08] W. Jo, K. Byung-Ki, **Kim Ji-Yeon**, H. Jung, K. Joo, T.-H. Oh, “The Devil is in the Details: Simple Remedies for Image-to-LiDAR Representation Learning.”
ACCV 2024
- [C07] **Kim Ji-Yeon**, O. Hyun-Bin, K. Byung-Ki, D. Kim, Y. Kwon, T.-H. Oh, “Depth-aware Video Panoptic Segmentation Model for Holistic Scene Understanding.”
Annual Conference of the Korean Robotics Society, 2024 (AFCV Best Robot Vision Paper Award)
- [C06] L. Hyoseok, K. Jun-Seong, **Kim Ji-Yeon**, T.-H. Oh “Feature Embedded 3D Gaussians for 3D Cognitive Map.”
Workshop on Image Processing and Image Understanding, 2024
- [C05] **Kim Ji-Yeon**, O. Hyun-Bin, D. Kim, T.-H. Oh “MinDVPS: Minimal Model for Depth-aware Video Panoptic Segmentation.”
CVPR 2023 Workshop on Vision Centric Autonomous Driving
- [C04] K. Youwang*, **Kim Ji-Yeon***, T.-H. Oh, “CLIP-Actor: Text-Driven Recommendation and Stylization for Animating Human Meshes.”
ECCV 2022 (QIFK Winner; AI4CC Workshop, CVPR 2023)
- [C03] **Kim Ji-Yeon**, T.-H. Oh, “Generating Action-Conditioned 3D Human in Scenes.”
Workshop on Image Processing and Image Understanding, 2022 (Outstanding Poster Award)

- [C02] K. Youwang, **Kim Ji-Yeon**, K. Joo, T.-H. Oh, "Unified 3D Mesh Recovery of Humans and Animals by Learning Animal Exercise."
BMVC 2021
- [C01] K. Youwang, **Kim Ji-Yeon**, K. Joo, T.-H. Oh, "Unified Pose and Shape Model for 3D Human-Animal Reconstruction."
Summer Annual Conference of the Institute of Electronics and Information Engineers, 2021

Patent

US 20240273798A1

Text-driven motion recommendation and neural mesh stylization system and a method for producing human mesh animation using the same

KR 10-2886014

Method and apparatus for motion animating and mesh stylization using text-driven motion recommendation

KR 10-2459293

Method and apparatus for generating mesh model of human or quadrupeds

Academic Service

Journal Reviewer

IEEE Robotics and Automation Letters (RA-L), 2024, 2026

The Visual Computer (TVCJ), 2022

Conference Reviewer

IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR), 2024 – 2025

IEEE/CVF International Conference on Computer Vision (ICCV), 2025

Conference on Neural Information Processing Systems (NeurIPS), 2024

The International Conference on Learning Representations (ICLR), 2024 – 2025, 2027

IEEE International Conference on Robotics and Automation (ICRA), 2023